SFU Beedie Business Analytics Hackathon 2026

One-Page Team Submission

GreenLeaf CEA · Precision Agriculture ROI Dashboard · Built in Tableau

TEAM ROSTER

Member	Program	Year	Relevant experience
Md Rownak Abtahee Diganta	BSc Honours Computing Science (Minor in Statistics)	4th	Statistics Minor, Machine Learning & Data Science Research Assistant, AI & Data Engineering Enthusiast, Cloud & AI Developer, Full-Stack AI Application Developer
Armaan Singh Chahal	BSc Computing Science (Minor in Statistics)	4th	Stats Minor, Cloud Computing Engineer, AWS AI pracitioner, AWS Cloud practitioner, AI research assistant
Kazi Boni Amin	BSc Computing Science	3rd	Tableau, Python, and applied analytics experience through CS and data projects. Built dashboards/KPI reports, analyzed user and performance data, and translated technical findings into clear business recommendations.

THE BUSINESS QUESTION

A B.C. farmer asks RBC for capital to expand a precision agriculture system. The lender needs evidence that sensor-driven decisions outperform routine management — not in theory, but in dollars per square metre. Our dashboard turns GreenLeaf's plot-level experimental data into that evidence, told in language a non-technical operator and a credit officer can both act on.

DASHBOARD CONCEPT — THREE LINKED VIEWS

One Tableau workbook, three views that follow the operator's day-to-night-to-season decision arc. Each view answers one question and hands off context to the next.

View	Decision it drives	Headline visuals	Key tables
1. Triage	Where should the crew go first this morning?	Ranked plot list by open alerts × stress × delay; per-plot drill-down timeline	daily_sensor, input_apps
2. ROI Story	Is the precision system worth financing? (RBC view)	Precision spend vs stress-reduction; precision_benefit_cad waterfall; Control vs treatment ROI	season_summary, daily_input_costs
3. Forecast	Which plots will fail next? (72-hr risk radar)	Plot-level risk score from VPD trend, canopy–air temp gap, and disease index; heatmap by greenhouse	daily_sensor, scouting

Differentiator: View 3 is forward-looking. Most teams will stop at retrospective ROI; we connect alert response time and substrate trend signals into a 72-hour at-risk plot list, framing precision tech as a *risk-management asset* — directly the language RBC uses when underwriting agricultural credit.

ANALYTIC APPROACH

- **Platform:** Tableau Desktop. Data prep in Tableau Prep (or pandas if needed). Single .twbx packaged for portability; no live connection dependency at presentation.
- **Data model:** Star schema with season_summary as the fact table at plot grain; daily_sensor_readings and daily_input_costs as time-series facts joined on plot_id + date. Treatment, crop, structure_type as filterable dimensions.
- **Key derived metrics:** Response Rate (action_taken ÷ alert_flag), Mean Response Time (action_delay_days on alert days), Precision $/Stress-Point Reduced (precision spend ÷ |post_action_stress_delta_3d|), and Precision Lift (treated plot profit − Control profit per crop).
- **Risk score (View 3):** Weighted index of 7-day rolling VPD, canopy–air temp divergence, disease_risk_index, and recent unresolved alerts. Tuned against plots that actually deteriorated in scouting notes.

WHAT COULD MISLEAD A FARMER (AND HOW WE GUARD AGAINST IT)

- **Selection bias:** Precision plots may not be comparable to Control plots if treatments differ. We compare within crop × treatment_block to isolate the precision effect.
- **Sample size on Control:** With 120 plots across 7 treatments, per-cell n is small. We will surface confidence intervals on ROI deltas, not just point estimates.
- **Cost attribution:** daily_input_costs ≠ season_summary totals (case explicitly notes this). View 2 reconciles both sources and flags discrepancies rather than hiding them.